

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286328

Kind Code

A1

Publication Date

September 11, 2025

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CABLE CONNECTOR ASSEMBLY HAVING CONNECTOR IDENTIFICATION MODULE

Abstract

A cable connector assembly includes a cable connector that includes a connector housing extending between a mating end and a cable end. The cable connector includes cable modules received in a cavity in a module stack. Each cable module includes a module housing holding signal contacts and cables terminated to the signal contacts. The cable connector includes a connector identification module received in the cavity configured to be mated with the mating connector assembly with the cable modules. The connector identification module includes an identification device held by an identification module housing. The identification device has unique identification data stored on the identification device. The identification device is operably coupled to the identification contact to transmit the unique identification data to the mating connector assembly.

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Family ID: 1000008475407

Appl. No.: 19/051303

Filed: February 12, 2025

Related U.S. Application Data

us-provisional-application US 63563838 20240311

Publication Classification

Int. Cl.: H01R13/66 (20060101); H01R12/62 (20110101); H01R13/6592 (20110101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims benefit to U.S. Provisional Application No. 63/563,838, filed 11 Mar. 2024, titled “CABLE CONNECTOR ASSEMBLY HAVING CONNECTOR IDENTIFICATION MODULE”, the subject matter of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] The subject matter herein relates generally to communication systems.

[0003] Cable assemblies are used to electrically connect various components within a system, such as a server or network system. The cable assemblies extend between the components, such as network, compute, storage, memory, or other components. The components may be arranged in shelves or trays of a network rack. Some systems may include components such as cards, such as line cards, daughter cards, mother boards, and the like. Each cable assembly includes multiple cables arranged in a bundle or array. Each cable forms a data channel. As the density of cable assemblies in the network system increase, it is desirable to know the locations of the components within the system, such as for service, repair or replacement.

[0004] A need remains for a method and device for identifying components and locations of components within a network system.

BRIEF DESCRIPTION OF THE INVENTION

[0005] In one embodiment, a cable connector assembly is provided and includes a cable connector that includes a connector housing extending between a mating end and a cable end. The connector housing has walls to define a cavity. The cable connector includes cable modules received in the cavity. Each cable module includes a module housing, signal contacts held by the module housing, and cables held by the module housing. The signal contacts are arranged at the mating end and are configured to be mated with a mating connector assembly. The cables are terminated to the signal contacts and extend from the cable end of the cable connector. The cables define data channels. The cable connector includes a connector identification module received in the cavity and is configured to be mated with the mating connector assembly. The connector identification module includes an identification module housing, identification contacts held by the identification module housing, and an identification device held by the identification module housing. The identification contacts are arranged at the mating end and are configured to be mated with the mating connector assembly. The identification device has unique identification data stored on the identification device. The identification device is operably coupled to the identification contact to transmit the unique identification data to the mating connector assembly.

[0006] In another embodiment, a connector identification module is provided and includes an identification module housing that includes a housing body extending between a front and a rear. The housing body includes a first side and a second side. The housing body includes a top and a bottom. The identification module housing includes an inner cavity at least partially surrounded by the housing body. The connector identification module includes an identification contact assembly received in the inner cavity and held by the housing body. The identification contact assembly includes a contact holder holding identification contacts. The identification contacts include mating ends that extend forward of the front of the housing body for mating with a mating connector assembly. The connector identification module includes an identification device received in the inner cavity and held by the housing body. The identification device has unique identification data stored on the identification device. The identification device is operably coupled to the identification contacts to transmit the unique identification data to the mating connector assembly.

[0007] In a further embodiment, a communication system is provided and includes a cartridge that has panels forming a chamber. The cartridge includes ports at a front of the cartridge. The communication system includes cable connector assemblies received in the ports for mating with mating connector assemblies. Each cable connector assembly includes a cable connector and a connector identification module. The cable connector includes a connector housing extending between a mating end and a cable end. The connector housing has walls that define a cavity. The cable connector includes cable modules received in the cavity. Each cable module includes a module housing, signal contacts held by the module housing, and cables held by the module housing. The signal contacts are arranged at the mating end and configured to be mated with the corresponding mating connector assembly. The cables are electrically connected to the signal contacts and extend from the cable end of the cable connector. The cables define data channels. The connector identification module is received in the cavity and configured to be mated with the mating connector assembly. The connector identification module includes an identification module housing, identification contacts held by the identification module housing, and an identification device held by the identification module housing. The identification contacts are arranged at the mating end and configured to be mated with the mating connector assembly. The identification device has unique identification data stored on the identification device. The identification device is operably coupled to the identification contact to transmit the unique identification data to the mating connector assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a front view of a communication system in accordance with an exemplary embodiment.

[0009] FIG. 2 is a front view of a portion of the communication system in accordance with an exemplary embodiment.

[0010] FIG. 3 illustrates a portion of the communication system showing two of the cartridges in accordance with an exemplary embodiment.

[0011] FIG. 4 is a perspective view of a portion of the communication system showing one of the mating connector assemblies poised for coupling to the corresponding connector assembly in accordance with an exemplary embodiment.

[0012] FIG. 5 is a front perspective view of one of the connector assemblies in accordance with an exemplary embodiment.

[0013] FIG. 6 is a front perspective view of one of the contact modules in accordance with an exemplary embodiment.

[0014] FIG. 7 is an enlarged view of the mating end of the contact module in accordance with an exemplary embodiment.

[0015] FIG. 8 is a perspective view of the identification contact module in accordance with an exemplary embodiment.

[0016] FIG. 9 is a perspective view of a portion of the identification contact module in accordance with an exemplary embodiment with a device shell of the identification contact module removed to illustrate internal components of the identification contact module.

[0017] FIG. 10 is a perspective view of a portion of the identification contact module in accordance with an exemplary embodiment with a housing of the identification contact module removed to illustrate internal components of the identification contact module.

[0018] FIG. 11 is a perspective view of a portion of the identification contact module in accordance with an exemplary embodiment showing the identification device.

[0019] FIG. 12 is a front perspective view of the cable connector showing the identification contact

module poised for loading into the cable connector in accordance with an exemplary embodiment. [0020] FIG. **13** is a rear perspective view of the cable connector showing the identification contact module loaded into the cable connector.

[0021] FIG. **14** is a perspective view of the identification contact module showing an electronic device of the identification device in accordance with an exemplary embodiment.

[0022] FIG. **15** is a rear perspective view of the cable connector showing the identification contact module shown in FIG. **14** loaded into the cable connector.

DETAILED DESCRIPTION OF THE INVENTION

[0023] FIG. **1** is a front view of a communication system **100** in accordance with an exemplary embodiment. FIG. **2** is a front view of a portion of the communication system **100** in accordance with an exemplary embodiment. In an exemplary embodiment, the communication system **100** includes a chassis or rack **50** having a rack frame **52** holding electrical components **60**. The rack **50** may be a server rack. The electrical components **60** may include fixed modules **62** (FIG. **2**) and/or removable modules **64** (FIG. **1**). The removable modules **64** may be mated to the fixed modules **62**. For example, the removable modules **64** may be plugged into the rack frame **52** and removable from the rack frame **52**. In various embodiments, the electrical components **60** may include one or more patch panels, switches, servers, routers, firewalls, and the like. The rack frame **52** may hold a power supply, cooling components, and the like.

[0024] In an exemplary embodiment, the communication system **100** includes at least one connector assembly **102** (FIG. **2**) and at least one mating connector assembly **104** (FIG. **4**) configured to be mated with the corresponding at least one connector assembly **102**. The connector assembly **102** may be part of one of the electrical components **60**, such as one of the fixed modules **62**. The mating connector assembly **104** may be part of one of the electrical components **60**, such as one of the removable modules **64**. In various embodiments, each mating connector assembly **104** may include one or more mating electrical connectors mounted to a circuit board, such as a backplane, a daughter card, a network switch, and the like.

[0025] In an exemplary embodiment, the communication system **100** includes a connector identification system **200** to identify the various connector assemblies **102** and/or mating connector assemblies **104**. The connector identification system **200** may be used to identify physical locations of the connector assemblies **102** within the communication system **100**, such as to identify which connector in a cable harness is electrically connected to which electrical component **60** for system management. In an exemplary embodiment, the connector identification system **200** may use identification devices, such as EEPROMs (Electrically Erasable and Programmable Read Only Memory), to hold information that allows identification of the various connector assemblies **102**. The identification devices may be incorporated with each of the connector assemblies **102**. For example, the identification devices may be incorporated into the connector assembly **102**, such as housed within the connector assembly and having a mating interface configured to mate with the mating connector assembly **104**. The identification devices may have unique identification information programmed into the identification devices, such as a part number, a program name, a slot ID number, data about the associated connector assembly **102**, and the like. In an exemplary embodiment, the identification devices are modular and may be incorporated into different types of connector assemblies **102**.

[0026] In an exemplary embodiment, the communication system **100** includes a cabinet or cartridge **110** that holds the connector assemblies **102**. The connector assemblies **102** may be cable connector assemblies having cables extending from corresponding cable connectors. The cartridge **110** may form part of the fixed modules **62**. Optionally, multiple cartridges **110** may be provided (for example, left side and right side). The cartridge **110** may be coupled to the rack frame **52**, such as at the rear of the rack frame **52**. The cartridge **110** forms an enclosed space or chamber **111** for the connector assemblies **102** and the cables of the connector assemblies **102**. The cartridge **110** is used for cable management. For example, the cables may be part of cable harnesses routed within the

enclosed chamber **111** formed by the cartridge **110** electrically connecting the various connector assemblies **102**. The cartridge **110** is used to support or position the connector assemblies **102** for mating with the mating connector assemblies **104**.

[0027] FIG. **3** illustrates a portion of the communication system **100** showing two of the cartridges **110** in accordance with an exemplary embodiment. Each cartridge **110** holds a plurality of the connector assemblies **102**, such as in one or more rows and one or more columns. The connector assemblies **102** are provided at the front of the cartridge **110** for interfacing with the mating connector assemblies **104** (shown in FIG. **4**).

[0028] In an exemplary embodiment, the cartridge **110** is formed from a plurality of panels **112**, such as sheet metal panels. The cartridge **110** includes a front panel **114**, side panels **116**, and a rear panel **118**. The panels **112** may additionally include an upper panel and/or a lower panel. The connector assemblies **102** are provided at the front panel **114** for mating with the mating connector assemblies **104**. In an exemplary embodiment, the connector assemblies **102** are arranged in a column along the front panel **114**. Optionally, the connector assemblies **102** may be provided in multiple columns. The mating ends of the connector assemblies **102** are external to the cartridge **110**. For example, the connector assemblies **102** pass through ports or openings **120** in the front panel **114**. The connector assemblies **102** are coupled to the front panel **114** adjacent the openings **120**. In an exemplary embodiment, the connector assemblies **102** may float or move relative to the front panel **114** to align with and mate to the mating connector assemblies **104**. In an exemplary embodiment, the connector assemblies **102** include mating guides, such as guide posts **122**, to guide mating of the mating electrical connectors **106** with the connector assemblies **102**. The guide posts **122** extend forward of the front panel **114**.

[0029] FIG. **4** is a perspective view of a portion of the communication system **100** showing one of the mating connector assemblies **104** poised for coupling to the corresponding connector assembly **102**. The connector assembly **102** is coupled to the cartridge **110**, such as to the front panel **114**. In the illustrated embodiment, the mating connector assembly **104** is a network component. For example, the mating connector assembly **104** may be a compute node, a memory module, a network switch, a storage device, a network accelerator, or another type of communication component. The mating connector assembly **104** may be a backplane component or a daughtercard component.

[0030] In an exemplary embodiment, the mating connector assembly **104** includes a mating electrical connector **106** mounted to a circuit board **108**. The mating connector assembly **104** may be mounted to a front edge of the circuit board **108**. The mating electrical connector **106** includes a connector housing **109** holding signal contacts (not shown). The signal contacts are terminated to the circuit board **108**. The signal contacts may be socket contacts. The mating electrical connector **106** may include shields providing shielding for the signal contacts.

[0031] The connector assembly **102** is received in the cartridge **110**. The connector assembly **102** is provided at the front panel **114** for mating with the mating connector assembly **104** along a mating axis **126**. The mating end of the cable connector **130** is configured to pass through an opening in the front panel **114** for mating with the mating electrical connector **106**. The connector assemblies **102**, **104** are configured to transmit and/or receive data through an interface. In an exemplary embodiment, the identification devices of the connector identification system **200** are contained within the connector assemblies **102** to mate with the mating connector assemblies **104**. The connector assemblies **102** may include receptacle connectors and the mating connector assemblies **104** may include plug connectors. In an exemplary embodiment, the guide post **122** is configured to be mated with a guide module **124** of the mating connector assembly **104** to guide mating of the mating connector assembly **104** with the connector assembly **102**. In the illustrated embodiment, the guide module **124** includes an opening that receives the guide post **122**. The mating guides may be used to align the connectors in one or more lateral directions.

[0032] FIG. **5** is a front perspective view of one of the connector assemblies **102** in accordance

with an exemplary embodiment. In an exemplary embodiment, the connector assembly **102** includes a cable connector **130** having a connector identification module **202**. The connector identification module **202** is part of the connector identification system **200** used to identify the connector assembly **102** within the system. The connector identification module **202** is integrated into the cable connector **130**. The connector identification module **202** is configured to be mated to the mating connector assembly **104** with the cable connector **130**.

[0033] In an exemplary embodiment, the connector identification module **202** includes an identification contact module **204** configured to be plugged into the cable connector **130**. The identification contact module **204** includes an identification device **210** holding unique identification data. The identification device **210** is able to uniquely identify the cable connector **130**. For example, the identification device **210** is able to identify the cable connector **130** from a plurality of cable connectors, such as within the cartridge **110**. The identification device **210** may identify the location of the cable connector **130** or other identifying information about the cable connector **130**. In an exemplary embodiment, the identification device **210** is an EEPROM device that is programmable and configured to store unique identification data.

[0034] The cable connector **130** includes a connector housing **132** holding the identification contact module **204** and a plurality of communication contact modules **160** configured to transmit data, such as high-speed data signals. The contact modules **160** include contact assemblies **140** configured to be mated with the mating connector assembly **104**. The contact assemblies **140** are terminated to ends of cables **150** extending from the cable connector **130**. For example, the cables **150** may extend from the rear ends of the contact modules **160**. The contact assemblies **140** are arranged together within the contact modules **160**. For example, the contact modules **160** hold the contact assemblies **140** in columns. A plurality of the contact modules **160** are received in the connector housing **132**, such as being arranged in a contact module stack. The identification contact module **204** may be arranged with the contact modules **160** in the stack. For example, the identification contact module **204** and the contact modules **160** may be arranged side-by-side. In an exemplary embodiment, the identification contact module **204** is arranged in the contact module stack at one of the ends of the contact module stack.

[0035] The connector housing **132** includes a cavity **134** that receives the contact modules **160** and the identification contact module **204**. The contact assemblies **140** are arranged in the cavity **134** at a mating end for mating with the mating electrical connector **106**. For example, the contact assemblies **140** are arranged in the cavity **134** in rows and columns. The identification contact module **204** may include contact assemblies similar to the contact assemblies **140** to similarly mate with the mating electrical connector **106**. The cavity **134** may form a receptacle that receives the mating end of the mating electrical connector **106**. The walls of the connector housing **132** may be chamfered and have a lead-in surface to guide mating of the mating electrical connector **106** in the cavity **134**. The connector housing **132** may have guide features to properly position the mating electrical connector **106** within the cavity **134**. In an exemplary embodiment, the connector housing **132** includes a plurality of module slots **136** in the cavity **134**. Each module slot **136** receives one of the contact modules **160** or the identification contact module **204**. The contact modules **160** and the identification contact module **204** may be latchably coupled to the connector housing **132** in the corresponding module slots **136**. In an exemplary embodiment, the contact modules **160** and the identification contact module **204** have similar dimensions and features to be interchangeably received in the module slots **136**. For example, the identification contact module **204** may have the same exterior dimensions as the contact modules **160** such that the identification contact module **204** may be plugged into any of the module slots **136** without need to redesign the connector housing **132** to accommodate the identification contact module **204**.

[0036] With additional reference to FIG. **6**, which is a front perspective view of one of the contact modules **160** in accordance with an exemplary embodiment, and FIG. **7**, which is an enlarged view of the mating end of the contact module **160**, the contact module **160** includes a structure holding

the contact assemblies **140** and the cables **150**. For example, the contact module **160** includes a body **162** holding the contact assemblies **140** and the cables **150**. The cables **150** extend from a cable end of the contact module **160**. The contact assemblies **140** extend from a mating end of the contact module **160**.

[0037] The body **162** may be generally rectangular having a top, a bottom, a front, a rear, a first side and a second side. The sides may be generally planar and parallel to each other configured to be stacked in the module stack side-by-side with the other contact modules **160**. In an exemplary embodiment, the body **162** holds a ground shield **164**. The body **162** includes latches **166**, such as at the top and the bottom of the body **162**, used to secure the contact module **160** in the connector housing **132** (FIG. 5). The contact module **160** may include a strain relief **168**, such as at the cable end, configured to hold the ends of the cables **150** and provide strain relief for the cables **150**. The strain relief **168** may be an overmold body that is overmolded over the ends of the cables **150**.

[0038] Each contact assembly **140** includes at least one signal contact **142**, a contact holder **141** holding the at least one signal contact **142**, and a shield **144** coupled to the contact holder **141** and providing electrical shielding for the at least one signal contact **142**. Optionally, multiple contact assemblies **140** may be integrated into a common structure. For example, the contact holders **141** of each of the contact assemblies **140** may be a unitary structure, such as being a single molded piece holding all of the signal contacts **142** and the shields **144**.

[0039] In the illustrated embodiment, each contact assembly **140** includes a pair of the signal contacts **142**, which define a differential pair. In various embodiments, the signal contacts **142** may be deflectable spring beams. However, in alternative embodiments, the signal contacts **142** may be pin contacts, socket contacts, or other types of contacts. The signal contacts **142** may be stamped and formed contacts. Each signal contact **142** is configured to be electrically connected to a corresponding wire or conductor of the cable **150**. In an exemplary embodiment, the cable **150** is a twin-axial cable having a first conductor and a second conductor terminated to corresponding signal contacts **142**.

[0040] The contact holder **141** is manufactured from a dielectric material, such as a plastic material. The contact holder **141** may be a molded part. The contact holder **141** includes a contact channel **143** for each signal contact **142**. In the illustrated embodiment, the contact holder **141** has two contact channels **143** for the pair of signal contacts **142**. In various embodiments, the contact holder **141** is preformed and the signal contacts **142** are loaded into the contact holder **141**. In other various embodiments, the contact holder **141** is formed in situ around the signal contacts **142**, such as being overmolded over the signal contacts **142**. The cable **150** may extend into an end of the contact holder **141** to terminate to the signal contacts **142**. The contact holder **141** may provide strain relief for the end of the cable **150**. In other embodiments, a single contact holder may be provided for the contact module **160** configured to hold each of the pairs of the signal contacts **142**, such as four pairs of the signal contacts **142** in a single molded part.

[0041] The shield **144** may be a stamped and formed shield. The shield **144** is configured to be coupled to the contact holder **141**. Optionally, shields may be provided at both sides of the contact module **160**. The shield **144** may provide shielding on multiple sides of the signal contacts **142**. In an exemplary embodiment, the front end of the shield **144** is C-shaped providing electrical shielding on three sides of the pair of signal contacts **142**. The shield **144** may have other shapes in alternative embodiments. The shield **144** may have deflectable spring beams configured to be mated to a shield structure of the mating connector assembly **104**. The shield **144** may be configured to be electrically connected to the cable shield of the cable.

[0042] FIG. 8 is a perspective view of the identification contact module **204** in accordance with an exemplary embodiment. FIG. 9 is a perspective view of a portion of the identification contact module **204** in accordance with an exemplary embodiment with a device shell of the identification contact module **204** removed to illustrate internal components of the identification contact module **204**. FIG. 10 is a perspective view of a portion of the identification contact module **204** in

accordance with an exemplary embodiment with a housing of the identification contact module **204** removed to illustrate internal components of the identification contact module **204**. FIG. **11** is a perspective view of a portion of the identification contact module **204** in accordance with an exemplary embodiment showing the identification device **210**. The identification contact module **204** may be similar to the signal contact module **160** (shown in FIG. **7**). For example, the identification contact module **204** and the contact module **160** may have similar dimensions and components. The identification contact module **204** and the contact module **160** may be interchangeable within the contact module stack and arranged in the connector housing **132** for mating with the mating connector assembly **104**.

[0043] In an exemplary embodiment, the identification contact module **204** includes a plurality of the contact assemblies **240** configured to be mated with the mating connector assembly **104** and the identification device **210** operably coupled to the contact assemblies **240**.

[0044] In an exemplary embodiment, the identification device **210** includes an electronic device **212** that stores unique identification data. The electronic device **212** may be an integrated circuit, such as a chip. In an exemplary embodiment, the electronic device **212** is an EEPROM device, such as an EEPROM chip or component. The electronic device **212** is programmable to store the identification data, such as a part number, a program name, a slot ID number, and the like. The electronic device **212** may be a surface mount component, such as having solder tails, solder pads, solder balls, or other conductors or leads for connection to another component, such as a circuit board or directly to the contacts of the contact assemblies **240**.

[0045] In an exemplary embodiment, the identification device **210** includes a printed circuit board assembly **220** including a printed circuit board **222**, the electronic device **212** mounted to the printed circuit board **222**, and may include other components mounted to the printed circuit board **222**, such as capacitors, resistors, a processor, a memory module, or another component. The printed circuit board **222** includes pads **224** on one or more surfaces of the printed circuit board **222**. The pads **224** may be provided proximate to the edge, such as for connection to the contact assemblies **240**.

[0046] In an exemplary embodiment, the identification device **210** includes a device shell **230** (FIG. **8**) holding the printed circuit board assembly **220**. The device shell **230** surrounds the printed circuit board **222** and the electronic device **212**. In an exemplary embodiment, the device shell **230** includes an overmold body **232** overmolded over the printed circuit board assembly **220**. In alternative embodiments, the device shell **230** may include multiple shells coupled together to hold the printed circuit board assembly **220** therebetween. The device shell **230** surrounds the printed circuit board assembly **220** to protect the components, such as the electronic device **212**.

[0047] The identification contact module **204** includes a plurality of the identification contact assemblies **240**. The identification contact assemblies **240** may be similar to the contact assemblies **140** (shown in FIG. **7**). Each contact assembly **240** includes at least one identification contact **242**, a contact holder **241** holding the at least one identification contact **242**, and a shield **244** coupled to the contact holder **241** and providing electrical shielding for the at least one identification contact **242**. Optionally, multiple contact assemblies **240** may be integrated into a common structure. For example, the contact holders **241** of each of the contact assemblies **240** may be a unitary structure, such as being a single molded piece holding all of the identification contacts **242** and the shields **244**.

[0048] In various embodiments, the contact assembly **240** may include multiple identification contacts **242**. The identification contacts **242** may transmit power to the identification device **210**. The identification contacts **242** may be signal contacts, power contacts, ground contacts, or other types of contacts. In various embodiments, the identification contacts **242** may be deflectable spring beams. However, in alternative embodiments, the identification contacts **242** may be pin contacts, socket contacts, or other types of contacts. The identification contacts **242** may be stamped and formed contacts. Each identification contact **242** is configured to be electrically

connected to the identification device **210**, such as being terminated to the printed circuit board **222**.

[0049] The contact holder **241** is manufactured from a dielectric material, such as a plastic material. The contact holder **241** may be a molded part. The contact holder **241** includes a contact channel **243** for each identification contact **242**. In the illustrated embodiment, the contact holder **241** has two contact channels **243** for the pair of identification contacts **242**. In various embodiments, the contact holder **241** is preformed and the identification contacts **242** are loaded into the contact holder **241**. In other various embodiments, the contact holder **241** is formed in situ around the identification contacts **242**, such as being overmolded over the identification contacts **242**. In other embodiments, a single contact holder may be provided for the contact module **204** configured to hold each of the pairs of the identification contacts **242**, such as three pairs of the identification contacts **242** in a single molded part.

[0050] The shield **244** may be a stamped and formed shield. The shield **244** is configured to be coupled to the contact holder **241**. Optionally, shields may be provided at both sides of the contact module **204**. The shield **244** may provide shielding on multiple sides of the identification contacts **242**. In an exemplary embodiment, the front end of the shield **244** is C-shaped providing electrical shielding on three sides of the pair of identification contacts **242**. The shield **244** may have other shapes in alternative embodiments. The shield **244** may have deflectable spring beams configured to be mated to a shield structure of the mating connector assembly **104**.

[0051] In an exemplary embodiment, the identification contact module **204** includes an identification module housing **260** configured to hold the identification contact assemblies **240** and the identification device **210** and configured to be coupled to the connector housing **132** of the cable connector **130**. The identification module housing **260** includes a housing body **262** extending between a front **264** and a rear **266**. The housing body **262** includes a first side **270** and a second side **272**. The housing body **262** includes a top **274** and a bottom **276**. The identification module housing **260** includes an inner cavity **278** at least partially surrounded by the housing body **262**.

[0052] The identification contact assemblies **240** are received in the inner cavity **278** and held by the housing body **262**. For example, the contact holder(s) **241** may be coupled to the housing body **262**. In an exemplary embodiment, the contact assemblies **240** are provided at the front **264**. The contact holders **241** and the mating ends of the identification contacts **242** may extend forward of the front **264**, such as for mating with the mating connector assembly **104**.

[0053] The identification device **210** is received in the inner cavity **278** and held by the housing body **262**. For example, the housing body **262** may include posts **279** configured to hold the printed circuit board **222**. The posts **279** may be heat staked to secure the printed circuit board **222** in the inner cavity **278**. Other securing features may be used in alternative embodiments, such as latches, clips, fasteners, and the like.

[0054] In an exemplary embodiment, the housing body **262** includes an opening **280** at the first side **270** open to the inner cavity **278**. The contact assemblies **240** and the identification device **210** may be received in the inner cavity **278** through the opening **280**. In an exemplary embodiment, at least a portion of the opening **280** may be closed by the device shell **230** (FIG. 8). For example, the device shell **230** may be attached to the identification device **210**, such as to the printed circuit board **222**. The device shell **230** may be formed in place in the inner cavity **278**. For example, the device shell **230** may be an overmold or hot melt at least partially filling the inner cavity **278**. In various embodiments, the device shell **230** may be flush with the first side **270**.

[0055] In an exemplary embodiment, the housing body **262** may be generally rectangular. The first and second sides **270**, **272** may be generally planar and parallel to each other configured to be stacked in the module stack side-by-side with the other contact modules **160**. In an exemplary embodiment, the identification device **210** is contained within the inner cavity **278** of the housing body **262**. The identification device **210** is contained between the front **264** and the rear **266**,

between the top **274** and the bottom **276**, and between the first side **270** and the second side **272**. For example, the identification device **210** does not protrude from the sides allowing the identification contact module **204** to be stacked with the contact modules **160** in the cable connector **130**.

[0056] In an exemplary embodiment, the identification module housing **260** includes latches **284**, such as at the top **274** and the bottom **276** of the housing body **262**. The latches **284** are used to secure the identification contact module **204** in the connector housing **132** of the cable connector **130**.

[0057] FIG. **12** is a front perspective view of the cable connector **130** showing the identification contact module **204** poised for loading into the cable connector **130**. FIG. **13** is a rear perspective view of the cable connector **130** showing the identification contact module **204** loaded into the cable connector **130**. In an exemplary embodiment, the identification contact module **204** is rear loaded into the cavity **134**, such as into the corresponding module slot **136**. The identification contact module **204** is configured to be arranged in the module stack with the other contact modules **160**. The connector identification module **202** and the contact modules **160** may be interchangeably receivable in the contact assembly slots **136**. For example, because the connector identification module **202** and the contact modules **160** have the same dimensions and mating features, the connector identification module **202** may be arranged in any of the slots. When assembled, the contact assemblies **140** and the identification contact assemblies **240** are arranged at the mating end of the cable connector **130**, such as at the front of the cable connector **130**. The contact assemblies **140** and the identification contact assemblies **240** may be arranged in the receptacle for mating with the mating electrical connector **106**.

[0058] The identification device **210** of the identification contact module **204** is operably coupled to the identification contact assemblies **240**. The identification device **210** is configured to be electrically coupled to the mating electrical connector **106** by the identification contact assemblies **240**. The identification device **210** is used to identify the connector assembly **102** associated with the connector identification module **202**. The identification device **210** may be used to identify physical locations of the connector assembly **102** within the communication system **100**. In an exemplary embodiment, the identification device **210** may be an EEPROM. The identification device **210** holds information that allows identification of the connector assembly **102**. The identification device **210** may have unique identification information programmed into the identification device **210**, such as a part number, a program name, a slot ID number, and the like. The identification device **210** is operably coupled to the identification contacts **242** to transmit the unique identification data to the mating connector assembly **104**.

[0059] FIG. **14** is a perspective view of the identification contact module **204** showing an electronic device **212** of the identification device **210** in accordance with an exemplary embodiment. The electronic device **212** is an EEPROM device. The electronic device **212** includes leads **214** extending from the body of the electronic device **212**. The leads **214** are configured to be terminated directly to the corresponding identification contacts **242**. The identification device **210** is provided without a printed circuit board. The electronic device **212** may be coupled to the housing body **262** of the identification module housing **260**.

[0060] FIG. **15** is a rear perspective view of the cable connector **130** showing the identification contact module **204** shown in FIG. **14** loaded into the cable connector **130**. In an exemplary embodiment, the identification contact module **204** is rear loaded into the cavity **134**, such as into the corresponding module slot **136**. The identification contact module **204** is configured to be arranged in the module stack with the other contact modules **160**. The identification device **210** of the identification contact module **204** is operably coupled to the identification contact assemblies **240**. The identification device **210** is configured to be electrically coupled to the mating electrical connector **106** by the identification contact assemblies **240**. The identification device **210** is used to identify the connector assembly **102** associated with the connector identification module **202**. The

identification device **210** may be used to identify physical locations of the connector assembly **102** within the communication system **100**. In an exemplary embodiment, the identification device **210** may be an EEPROM. The identification device **210** holds information that allows identification of the connector assembly **102**. The identification device **210** may have unique identification information programmed into the identification device **210**, such as a part number, a program name, a slot ID number, and the like. The identification device **210** is operably coupled to the identification contacts **242** to transmit the unique identification data to the mating connector assembly **104**.

[0061] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope.

Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. A cable connector assembly comprising: a cable connector including a connector housing extending between a mating end and a cable end, the connector housing having walls defining a cavity, the cable connector including cable modules received in the cavity, each cable module including a module housing, signal contacts held by the module housing, and cables held by the module housing, the signal contacts arranged at the mating end and configured to be mated with a mating connector assembly, the cables terminated to the signal contacts and extending from the cable end of the cable connector, the cables defining data channels; and a connector identification module received in the cavity and configured to be mated with the mating connector assembly, the connector identification module including an identification module housing, identification contacts held by the identification module housing, and an identification device held by the identification module housing, the identification contacts arranged at the mating end and configured to be mated with the mating connector assembly, the identification device having unique identification data stored on the identification device, the identification device being operably coupled to the identification contact to transmit the unique identification data to the mating connector assembly.
2. The cable connector assembly of claim 1, wherein the cable modules and the connector identification module are arranged in a module stack side-by-side in the cavity of the connector housing.
3. The cable connector assembly of claim 1, wherein the connector housing includes housing slots in the cavity, the module housings and the identification module housing have similar dimensions configured to be interchangeably received in the housing slots.
4. The cable connector assembly of claim 1, wherein the identification device includes a printed circuit board assembly including a printed circuit board and an electronic device mounted to the

printed circuit board, the unique identification data being stored on the electronic device, the identification contacts being terminated to the printed circuit board.

5. The cable connector assembly of claim 1, wherein the identification device is an EEPROM device.

6. The cable connector assembly of claim 1, wherein the connector identification module includes a device shell surrounding the identification device.

7. The cable connector assembly of claim 1, wherein the connector identification module includes shields providing shielding for the identification contacts.

8. The cable connector assembly of claim 1, wherein each module housing includes latches being latchably coupled to the connector housing to secure the cable module in the cavity, the identification module housing including latches being latchably coupled to the connector housing to secure the connector identification module in the cavity.

9. The cable connector assembly of claim 1, wherein the connector identification module includes contact holders holding the identification contacts in pairs, the connector identification module including shields coupled to the contact holders to provide shielding for the identification contacts, the identification device includes a printed circuit board assembly including a printed circuit board and an electronic device mounted to the printed circuit board, the unique identification data being stored on the electronic device, the contact holders located forward of the printed circuit board, the identification contacts and the shields being terminated to the printed circuit board.

10. A connector identification module comprising: identification module housing including a housing body extending between a front and a rear, the housing body including a first side and a second side, the housing body including a top and a bottom, the identification module housing including an inner cavity at least partially surrounded by the housing body; an identification contact assembly received in the inner cavity and held by the housing body, the identification contact assembly including a contact holder holding identification contacts, the identification contacts including mating ends extending forward of the front of the housing body for mating with a mating connector assembly; and an identification device received in the inner cavity and held by the housing body, the identification device having unique identification data stored on the identification device, the identification device being operably coupled to the identification contacts to transmit the unique identification data to the mating connector assembly.

11. The connector identification module of claim 10, wherein the identification device includes a printed circuit board assembly including a printed circuit board and an electronic device mounted to the printed circuit board, the unique identification data being stored on the electronic device, the identification contacts being terminated to the printed circuit board.

12. The connector identification module of claim 10, wherein the identification device is an EEPROM device.

13. The connector identification module of claim 10, wherein the identification device includes an electronic device storing the unique identification data and a device shell surrounding the electronic device, the device shell at least partially filling the inner cavity.

14. The connector identification module of claim 10, wherein the identification contact assembly includes shields providing shielding for the identification contacts.

15. The connector identification module of claim 10, wherein the identification module housing including latches at the top and the bottom of the housing body configured to be latchably coupled to a connector housing of a connector assembly.

16. The connector identification module of claim 10, wherein the housing body includes an opening at the first side open to the inner cavity, the contact assembly and the identification device being received in the inner cavity through the opening.

17. The connector identification module of claim 10, wherein the identification device is contained within the inner cavity of the housing body between the front and the rear, between the top and the bottom, and between the first and second sides.

18. A communication system comprising: a cartridge having panels forming a chamber, the cartridge including ports at a front of the cartridge; cable connector assemblies received in the ports for mating with mating connector assemblies, each cable connector assembly including a cable connector and a connector identification module, the cable connector including a connector housing extending between a mating end and a cable end, the connector housing having walls defining a cavity, the cable connector including cable modules received in the cavity, each cable module including a module housing, signal contacts held by the module housing, and cables held by the module housing, the signal contacts arranged at the mating end and configured to be mated with the corresponding mating connector assembly, the cables electrically connected to the signal contacts and extending from the cable end of the cable connector, the cables defining data channels; and the connector identification module received in the cavity and configured to be mated with the mating connector assembly, the connector identification module including an identification module housing, identification contacts held by the identification module housing, and an identification device held by the identification module housing, the identification contacts arranged at the mating end and configured to be mated with the mating connector assembly, the identification device having unique identification data stored on the identification device, the identification device being operably coupled to the identification contact to transmit the unique identification data to the mating connector assembly.

19. The communication system of claim 18, wherein the connector identification modules uniquely identify the positions of the corresponding cable connector assemblies within the communication system based on the unique identification data.

20. The communication system of claim 18, wherein the identification device is an EEPROM device.
